Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Part 1: Course Information

Course: ISE 218: Fundamentals of Information Technology

Semester: Spring 2025

Main Instructor name: Abid Malik

Co-Instructor name: Xinrui Yu

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Course hours: Friday (09:50 AM - 12:15 PM)

Office hours: Monday 9:00 AM - 10:00 AM, Friday 08:30 AM - 09:30 AM, Lixi Building 201-4

Prerequisites:

Course Description:

This course introduces computer organization to students interested in the relationship between computer hardware and information systems. The course examines components found in high use computing devices such as desktop computers, smart phones and navigation systems. The focus of the examination is understanding the underlying technology of each component, along with price/performance curves and competing technologies. Upon completion of the course, student should be proficient in reading device specifications, particularly the functional and performance implications. Students should also be able to use that knowledge to compare competing devices.

Major Topics Covered:

- Data representation and compression
- Fundamentals of computer architecture
- Main memory technologies
- Input/output and storage devices of operating systems internals
- System performance measurement
- Basics of computer networks

Required Course Textbook and Materials:

Required: *Essentials of Computer Organization and Architecture (5th edition)* by Linda Null and Julia Lobur. Jones & Bartlett Learning, 2018. ISBN 9781284123036

Course Delivery:

Lixi Building 314

How We Will Communicate:

Email: amalik@cs.stonybrook.edu xryu47@gmail.com

Part 2: Course Learning Objectives

Upon completion of the course, students will be able to:

1) Understand how numerical and non-numerical data are represented in digital computers,
2) Understand the functions performed by components of a computer-based device,
3) Indicate strengths and weaknesses inherent in different components and different architectures, and
4) Understand the specification sheet of a typical computer-based device

Part 3: Course Schedule

The Course Schedule is subject to change.

Week #	Date	Topic
Week 1	02/21	• Course Overview • Unit 1: Hardware Basics **• Homework #1 out (due Feb 28)**
Week 2	02/28	• Unit 2: Data Representation Read Sections 2.1 - 2.4.4, 2.4.6, 2.5.4-2.5.5 **• Homework #2 out (due March 07)**
Week 3	03/07	• Unit 3: Basics of Computer Architecture
Week 4	03/14	• Unit 3: Basics of Computer Architecture **• Homework 3 out (due March 28)** • Review Quiz 1
Week 5	03/21	**• Quiz 1 - Unit 1 and 2** • Unit 4: Instruction Set Architecture
Week 6	03/28	• Unit 4: Instruction Set Architecture **• Homework 4 out (due April 11)**
Week 7	04/04	04/04: Qingming Jie holiday
Week 8	04/11	• Unit 5 Memory **• Homework 5 out (due April 25)** **• Quiz 2 - Unit 3**
Week 9	04/18	**• Exam 1 (Unit 1,2,3,4)**
Week 10	04/25	• Unit 5: Memory

Week 11	05/02	05/02: Labor Day holiday
Week 12	05/09	• Unit 6: Input/Output Systems
Week 13	05/16	• Unit 6: Input/Output Systems • **Homework 6 out (due June 06)** • Unit 7: System Software
Week 14	05/23	• Unit 7: System Software
Week 15	05/30	**• Exam 2 (Unit 5,6)**
Week 16	06/06	• Unit 7: System Software • **Homework 7 out (due June 20)** • Unit 8: Network Organization and Architecture
Week 17	06/13	**• Quiz 3 (Unit 7)** • Unit 8: Network Organization and Architecture
Week 18	06/20	• Unit 9: Performance Measurement and Analysis Review Exam 3
Week 19	06/27	**06/23-07/02: Final - Exam 3 (Unit 7,8,9)**

Part 4: Grading, Attendance, and Late Work Policies

In this course, you will be assessed on the following:

Composition of Grade	Percentage	Dates
Homework Assignments	20	• February 28 • March 07 • April 11 • April 25 • June 06 • June 20
Quizzes	20	• March 21 • April 11 • June 13
Exam 1	20	• April 18
Exam 2	20	• May 30
Exam 3	20	• TBD

- Attendance and Late Policy: Absence or lateness will affect your grade.
- Assignments, Tests and Quizzes: Due dates will be announced in class.
- Late Work Policy: ALL WORK MUST BE SUBMITTED N OR BEFORE THE DUE DATE AND TIME.
  - Assignments submitted late will lose 5 points per day.
  - Incomplete assignments or assignments that do not fulfill requirements will be returned and not counted.

Part 5: Academic Integrity Statement

- Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Any form of academic dishonesty, including cheating and plagiarism, will be reported to the Institute.